QryptoNox: Quantum Oracle Forensics

Adaptive Identification of Hidden Oracle Families

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Abstract

Quantum oracle forensics is the task of identifying an unknown oracle from its input-output behaviour under a limited query budget. This project develops an adaptive solver that classifies hidden quantum-oracle families and recovers associated secret information where required. The solver considers eight families: constant, balanced, linear, Grover, random-function, Simon, Simon2, and permutation oracles. The implementation is evaluated under oracle misfire and measurement readout noise. In an initial experiment of 80 instances, the solver achieved an overall accuracy of 91.25%, with strong results on the Simon and permutation families. The results also reveal that distinguishing balanced, random-function, and Grover cases remains challenging. This report describes the approach, preliminary evaluation, limitations, and experiments needed for a more complete assessment.

1 Problem Formulation

An oracle is a black-box operation whose internal structure is unknown to the solver. Given access to an oracle and a restricted query budget, the objective is to infer its family and, where applicable, identify its hidden secret. This setting combines quantum algorithm design, statistical decision-making, and resource-aware experimentation.

For a Boolean function f, a standard quantum oracle acts as

$$U_f|x, y\rangle = |x, y \oplus f(x)\rangle,$$

where x denotes the input, y is the target bit, and $\oplus$ denotes XOR. By preparing the target in an appropriate superposition, an algorithm can exploit phase kickback and interference to obtain information about f.

The solver must return a prediction for the oracle family and its associated secret, subject to the challenge interface. It must balance classification accuracy against a finite query budget. Oracle misfires and readout errors make the task more difficult because observed outcomes may differ from ideal algorithmic predictions.

The evaluated families are constant, balanced, linear, Grover, random-function (randfun), Simon, Simon2, and permutation (perm). The challenge configuration uses $n = 8$, with $m = 1$ for constant, balanced, linear, Grover, and random-function cases, and $m = 8$ for Simon, Simon2, and permutation cases. The maximum configured budget is 60 oracle gate applications multiplied by shots, according to the challenge evaluator.

2 Methodology

2.1 Adaptive decision strategy

The solver uses a family-dependent decision strategy rather than applying one quantum algorithm to every oracle. This is motivated by the fact that different oracle families expose different mathematical structures. The implementation first uses observed outcomes to guide its family-specific procedure, then constructs the final family and secret prediction according to the solver's decision logic.

The main algorithmic ideas underlying the family-specific branches are:

- **Constant and balanced:** use oracle response behaviour to distinguish the two functional classes.
- **Linear:** exploit structured relationships between input and output information to infer the hidden linear behaviour.
- **Grover:** use search-oriented behaviour to identify a marked candidate and verify the candidate before accepting it.
- **Random-function:** use observed outcomes to distinguish unstructured behaviour from the more structured families.
- **Simon and Simon2:** exploit relationships between oracle outputs and hidden XOR structure to infer the secret.
- **Permutation:** use the observed oracle behaviour to identify the relevant permutation-family secret.

The implementation includes stochastic behaviour and verification logic where applicable. Such mechanisms can reduce the impact of unreliable individual observations, but they do not guarantee correct classification. In particular, different oracle families may produce overlapping observations when the number of queries is small or when noise is present.

2.2 Noise model and resource constraints

The initial evaluation uses an oracle misfire probability of 0.04 and a readout-flip probability of 0.02. A misfire can corrupt an oracle interaction, while readout noise can change the observed measurement outcome. The solver operates under a maximum budget of 60, making query allocation an important design consideration.

The evaluation interface also permits changing the random seed, number of trials, budget, misfire probability, and readout probability. These controls make it possible to test reproducibility and robustness under different conditions. However, the results reported here cover only the initial configuration; a systematic sweep across query budgets and increased noise remains future work.

3 Experimental Setup

The initial baseline experiment used the frozen solver with the following configuration:

Parameter	Value
Number of oracle families	8
Trials per family	10
Total evaluated instances	80
Random seed	20261019
Maximum query budget	60
Oracle misfire probability	0.04
Readout-flip probability	0.02

Each family contributes ten instances, so the overall accuracy is calculated across 80 predictions. Family-level accuracy is the fraction of correctly classified instances within that family. The average query count is reported per family by the evaluator. Since each family has the same number of trials, the overall mean query count is the arithmetic mean of these eight family averages.

4 Results and Discussion

4.1 Preliminary classification results

Table 1 summarizes the initial evaluation.

Table 1: Initial results under the baseline noise configuration.

Family	Accuracy	Average queries
Constant	100%	31.2
Balanced	80%	32.0
Linear	100%	16.0
Grover	80%	42.0
Random-function	70%	32.0
Simon	100%	50.0
Simon2	100%	50.0
Permutation	100%	50.0
Overall	91.25%	37.9

The solver correctly classified 73 out of 80 instances, corresponding to an overall accuracy of $73/80 = 91.25\%$. Constant, linear, Simon, Simon2, and permutation cases achieved 100% accuracy in this small evaluation. Balanced and Grover cases each achieved 80%, while random-function cases achieved 70%.

The overall mean query count was 37.9. The linear family had the lowest average query count at 16, whereas Simon, Simon2, and permutation each averaged 50 queries. This suggests that the current solver's resource use

depends substantially on the family-specific procedure. The query averages should not be interpreted as evidence of optimal query complexity: the present experiment does not compare them against proven lower bounds or a comprehensive set of alternative strategies.

4.2 Error analysis

The observed confusion patterns provide useful information about where the classifier fails. Of the ten balanced instances, eight were classified correctly and two were labelled random-function. Of the ten random-function instances, seven were correctly classified and three were labelled balanced. This reciprocal confusion indicates that these two families are difficult to distinguish using the current observations and decision rules.

For the Grover family, eight of ten instances were classified correctly. One was predicted as random-function and one as constant. These errors suggest that the solver's current evidence for Grover behaviour is not sufficiently decisive in every noisy instance. Verification and more informative query allocation may help, although their impact must be measured experimentally.

The perfect scores for several families are encouraging but should be interpreted cautiously. Each family has only ten test instances, and the results are tied to one random seed and one noise setting. They do not establish that the solver will achieve the same accuracy on other instances or under more severe noise.

5 Limitations and Future Evaluation

The current results are preliminary. The following experiments are needed for a more complete evaluation:

1. **Query-budget sensitivity:** evaluate at caps of 20, 30, 40, 50, and 60, reporting overall and per-family accuracy.
2. **Noise robustness:** repeat the experiment at increased misfire and readout probabilities, including the challenge's specified stress configuration.
3. **Baseline comparisons:** compare against random guessing, a classical-only strategy, a single-algorithm strategy, and an adaptive-off configuration where supported by the evaluator.
4. **Repeated-seed evaluation:** run additional seeds and more trials per family to estimate variability and reduce dependence on a single sample.
5. **Efficiency analysis:** report query use alongside accuracy at each budget cap to assess the trade-off between resource consumption and performance.

These experiments are necessary before making strong claims about robustness, efficiency, or superiority over alternative solvers. In particular, the current report does not claim that the solver outperforms the required baselines, since those comparisons have not yet been completed.

6 Conclusion

QryptoNox presents an adaptive approach to identifying hidden quantum-oracle families under a limited query budget. In the initial evaluation, the frozen solver achieved 91.25% accuracy across 80 instances, with an overall mean query count of 37.9. The strongest observed results were obtained on constant, linear, Simon, Simon2, and permutation families. The principal errors occurred between balanced and random-function oracles, with additional confusion in Grover cases.

The experiment establishes a useful initial benchmark, not a complete robustness assessment. Further budget sweeps, stress tests, baseline comparisons, and repeated-seed experiments are needed to determine how consistently the solver performs under changing conditions.